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Ferroelectric device and wave computing device

Abstract

A ferroelectric device and a wave computing device are provided. The ferroelectric device includes a first electrode, a second electrode, a ferroelectric layer and a wave guide. The ferroelectric layer is disposed between the first and second electrodes, and configured to transduce an electrical wave signal to a varying mechanical stress by piezoelectricity, or vice versa. A first polarization state or a second polarization state opposite to the first polarization state is programmed in the ferroelectric layer. The wave guide is in contact with the ferroelectric layer, and configured to transmit a wave signal resulted from or resulting the varying mechanical stress. The wave signal is in phase with the electrical wave signal when the ferroelectric layer is programmed with the first polarization state. The wave signal is out of phase with the electrical wave signal when the ferroelectric layer is programmed with the second polarization state.

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Background/Summary

BACKGROUND
(1) Most computing systems rely on paradigms, in which data is represented by electric charge or voltage, and computation is performed by charge movements. A fundamental circuit element of this framework is a field effect transistor (FET), which can be functioned as a switch and/or an amplifier. A plurality of the FETs with complementary conductive types can be interconnected to form a complementary metal-oxide-semiconductor (CMOS) circuit that can be operated to perform various logic functions. For many years, scaling of the CMOS circuit has been accompanied by research on alternative computing paradigms beyond the CMOS horizon.
(2) Wave computing is one of the beyond-CMOS approaches. In a paradigm of wave computing, data is encoded in amplitude and/or phase of a wave, and computation is done by wave interference. Inversion of a signal is essential in wave computing. However, inversion of a wave

signal is typically implemented by propagating the wave along a delay line, or by interference with a reference wave. Both of these implementations cost valuable chip area and may be undesired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. **1**A and FIG. **1**B are schematic cross-sectional views illustrating a ferroelectric device at different polarization states, according to some embodiments of the present disclosure.

(3) FIG. **1**C and FIG. **1**D are schematic cross-sectional views illustrating a ferroelectric device being used to generate electrical wave signals in response to an input of a sound wave signal, according to some embodiments of the present disclosure.

(4) FIG. **2**A through FIG. **2**E illustrate various layout designs of a ferroelectric device, according to some embodiment of the present disclosure.

(5) FIG. **3** is a schematic cross-sectional view illustrating a ferroelectric device, according to some embodiments of the present disclosure.

(6) FIG. **4**A is a schematic cross-sectional view illustrating a ferroelectric device, according to some embodiments of the present disclosure.

(7) FIG. **4**B is a schematic cross-sectional view illustrating a ferroelectric device, according to some embodiments of the present disclosure.

(8) FIG. **4**C is a schematic cross-sectional view illustrating a ferroelectric device, according to some embodiments of the present disclosure.

(9) FIG. **4**D is a schematic cross-sectional view illustrating a ferroelectric device, according to some embodiments of the present disclosure.

(10) FIG. **5**A is a schematic cross-sectional view illustrating a wave computing logic device, according to some embodiments of the present disclosure.

(11) FIG. **5**B is a schematic cross-sectional view illustrating a wave computing logic device having ferroelectric devices pre-programmed with different polarization states, according to some embodiments of the present disclosure.

(12) FIG. **6**A is a schematic cross-sectional view illustrating a wave computing logic device, according to some embodiments of the present disclosure.

(13) FIG. **6**B is a schematic cross-sectional view illustrating a wave computing logic device having ferroelectric device pre-programmed with another polarization state, according to some embodiments of the present disclosure.

(14) FIG. **7**A is a schematic plan view illustrating a differential wave signal generator, according to some embodiments of the present disclosure.

(15) FIG. **7**B is a schematic cross-sectional view along X-X′ line shown in FIG. **7**A.

(16) FIG. **8** is a schematic diagram illustrating that the differential wave signal generator as shown in FIG. **7** is further connected to a wave computing logic device, according to some embodiments of the present disclosure.

(17) FIG. **9** is a schematic diagram illustrating an error correction device used in a wave computing circuit, according to some embodiments of the present disclosure.

(18) FIG. **10** is a schematic cross-sectional view illustrating a semiconductor die, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(19) The following disclosure provides many different embodiments or examples, for

implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(20) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(21) In the following, signals, or waves, or wave signals, being either “in phase” or “out of phase” refers to coherent waves having a phase difference of nominally zero degrees, or coherent waves having a phase difference of nominally 180° (or pi radials, or half a period), respectively.

(22) Ferroelectric devices used in a paradigm of wave computing are provided in various embodiments of the present disclosure. The ferroelectric devices are reconfigurable transducers (input actuators or output sensors), and can be interconnected to form a logic gate, a differential wave generator, an error correction device or other wave computing devices. Specifically, each ferroelectric device can either pass a wave signal or invert the wave signal (in terms of phase), depending on a pre-programmed polarization state of the ferroelectric device. When a ferroelectric device is configured to invert a wave signal, the wave signal can be inverted at the ferroelectric device, without propagation along a delay line, or interference with a reference wave.

(23) FIG. **1**A and FIG. **1**B are schematic cross-sectional views illustrating a ferroelectric device **100** at different polarization states P**1**, P**2**, according to some embodiments of the present disclosure.

(24) Referring to FIG. **1**A and FIG. **1**B, the ferroelectric device **100** as a wave transducer includes a pair of electrodes **102***a*, **102***b*, and includes a ferroelectric layer **104** as well as a wave guide **106** lying between the electrodes **102***a*, **102***b*. According to some embodiments, the ferroelectric device **100** is an input actuator, and the electrode **102***a* is configured to receive an electrical wave signal EW. The electrical wave signal EW is an AC voltage signal, such as a radio frequency (RF) voltage signal. On the other hand, the electrode **102***b* may be coupled to a reference voltage, such as a ground voltage.

(25) The ferroelectric layer **104** lies between the electrode **102***a* and the wave guide **106**, and is formed of a ferroelectric material. Due to ferroelectricity of the ferroelectric material, the ferroelectric layer **104** can possess reversible electric polarization. Further, the ferroelectricity may be related to atom displacement. As shown in FIG. **1**A, the ferroelectric layer **104** is programmed with a polarization state P**1**, and an atom X in a unit cell in of the ferroelectric layer **104** may be displaced upwardly from a lattice site. On the other hand, as shown in FIG. **1**B, the ferroelectric layer **104** is programmed with a polarization state P**2** opposite to the polarization state P**1** in terms of direction, and the atom X in the unit cell of the ferroelectric layer **104** is displaced downwardly from the lattice site. In either case, a lattice constant of the ferroelectric layer **104** changes with variation of an electric field applied across the ferroelectric layer **104**, such as a varying electric field provided by the electrical wave signal EW. Depending on the polarization states P**1**, P**2**, the lattice constant of the ferroelectric layer **104** may change in opposite ways with respect to variation

of the electric field applied across the ferroelectric layer **104**. As an example, when the ferroelectric layer **104** is programmed with the polarization state P**1**, the lattice constant of the ferroelectric layer **104** may increase/decrease as the electric field applied across the ferroelectric layer **104** also increases/decreases. In other words, the lattice constant variation of the ferroelectric layer **104** programmed with the polarization state P**1** may be in phase with the varying electric field. On the other hand, when the ferroelectric layer **104** is programmed with the polarization state P**2**, the lattice constant of the ferroelectric layer **104** may increase as the electric field applied across the ferroelectric layer **104** decreases, and may decrease as the electric field increases. In other words, the lattice constant variation of the ferroelectric layer **104** programmed with the polarization state P**2** may be 180° out of phase with respect to the varying electric field.

(26) The lattice constant variation of the ferroelectric layer **104** in response to the varying electric field provided by the electrical wave signal EW results in varying mechanical stress applied to the wave guide **106** extending between the ferroelectric layer **104** and the electrode **102***b*. In other words, an electrical energy is transduced to a mechanical energy by a piezoelectricity (inverse piezoelectricity) of the ferroelectric layer **104**. As a result of such transduction, a sound wave signal SW**1**/SW**2** created by collective atom displacement may be generated in the wave guide **106**, and travels along the wave guide **106**. When the ferroelectric layer **104** is programmed with the polarization state P**1**, the lattice constant variation of the ferroelectric layer **104** may be in phase with the varying electric field provided by the electrical wave signal EW, thus the sound wave signal SW**1** created in the wave guide **106** may be in phase with the varying electric field as well. On the other hand, when the ferroelectric layer **104** is programmed with the polarization state P**2**, the lattice constant variation of the ferroelectric layer **104** may be out of phase with respect to the varying electric field, then the sound wave signal SW**2** created in the wave guide **106** may also be out of phase with respect to the varying electric field. Therefore, by pre-programming the ferroelectric layer **104** with the polarization state P**2**, a signal provided to the ferroelectric layer **104** can be inverted by the ferroelectric layer **104**, and such inversion takes place without the signal further propagating along a delay line, or interference with an additional reference signal.

(27) Prior to the transduction, the ferroelectric layer **104** is programmed with the polarization state P**1** or the polarization state P**2**. During a programming process, a programming voltage (e.g., a DC voltage larger than a coercive voltage of the ferroelectric layer **104**) is provided across the ferroelectric layer **104**. Particularly, a polarity of the programming voltage used for inducing the polarization state P**1** is opposite to a polarity of the programming voltage used for inducing the polarization state P**2**. On the other hand, an amplitude of the programming voltage used for inducing the polarization state P**1** is identical or different from an amplitude of the programming voltage used for inducing the polarization state P**2**. Further, an amplitude of the electrical wave signal EW provided during transduction should be lower than the amplitude(s) of the programming voltages used for inducing the polarization states P**1**, P**2**, to avoid from accidentally altering the polarization state of the ferroelectric layer **104**.

(28) The ferroelectric layer **104** is formed of a ferroelectric material possessing ferroelectricity and piezoelectricity (including direct piezoelectricity and inverse piezoelectricity). The piezoelectricity (inverse piezoelectricity) is utilized to transduce an electrical energy to a mechanical energy, and the ferroelectricity is utilized to determine whether to invert (in terms of phase) the incoming electrical wave signal. It should be appreciated that, a ferroelectric material is also a piezoelectric material, but a piezoelectric material is not necessarily a ferroelectric material. As examples, the ferroelectric layer **104** may be formed of AlScN, orthorhombic hafnium oxide, orthorhombic hafnium zirconium oxide (HZO), perovskite lead titanate, perovskite barium titanate or another suitable ferroelectric material. On the other hand, in order to transmit the sound wave signal SW**1**/SW**2**, the wave guide **106** may be formed of a crystalline material. As examples, the wave guide **106** may be formed of quartz, sapphire, diamond or metal (e.g., Al, Be, Mo, Ti or the like).

(29) According to the above embodiments, the ferroelectric device **100** generates the sound wave

signal SW**1**/SW**2** in response to an input of the electrical wave signal EW. In other embodiments, the ferroelectric device **100** can be used to generate an electrical wave signal in response to a sound wave signal.

(30) FIG. **1**C and FIG. **1**D are schematic cross-sectional views illustrating the ferroelectric device **100** being used to generate electrical wave signals EW**1**, EW**2** in response to an input of a sound wave signal SW, according to some embodiments of the present disclosure.

(31) Referring to FIG. **1**C and FIG. **1**D, according to some embodiments, the ferroelectric device **100** is functioned as an output sensor, and the wave guide **106** is configured to receive a sound wave signal SW. Due to piezoelectricity (direct piezoelectricity) of the ferroelectric layer **104**, the ferroelectric layer **104** can generate an electrical wave signal EW**1**/EW**2** while being stressed by the mechanical stress resulted from the sound wave signal SW, and the electrical wave signal EW**1**/EW**2** may be output via the electrode **102***a*. On the other hand, the electrode **102***b* may be coupled to a reference voltage, such as a ground voltage.

(32) Further, phases of the electrical wave signals EW**1**, EW**2** are dependent on a pre-programmed polarization state of the ferroelectric layer **104**. As shown in FIG. **1**C, when the ferroelectric layer **104** is programmed with the polarization state P**1**, the generated electrical wave signal EW**1** is in phase with the sound wave signal SW. On the other hand, as shown in FIG. **1**D, the generated electrical wave signal EW**2** is out of phase with the sound wave signal SW when the ferroelectric layer **104** is programmed with the polarization state P**2**. Particularly, the phase inversion occurs right at the ferroelectric device **100**, without help of a delay line or an additional reference signal.

(33) FIG. **2**A through FIG. **2**E illustrate various layout designs of the ferroelectric device **100**, according to some embodiment of the present disclosure. It should be noted that, the ferroelectric layer **104** and the electrode **102***b* overlapped by the electrode **102***a* are not shown in FIG. **2**A through FIG. **2**E.

(34) Referring to FIG. **2**A, the wave guide **106** laterally protrudes from the electrode **102***a* and the underlying ferroelectric layer **104** (not shown). In some embodiments, the wave guide **106** laterally extends along a single direction X at a single side of a stacking structure including the electrode **102***a* and the underlying ferroelectric layer **104** (not shown). In these embodiments, a terminal of the wave guide **106** away from the stacking structure may be coupled to an input/output (I/O) of a single wave computing device, such as a logic gate.

(35) In the embodiments as shown in FIG. **2**A, the wave guide **106** has a single protruding portion. In following embodiments, the wave guide **106** has multiple protruding portions coupled to multiple wave computing devices, such as logic gates. The sound wave signal generated by the ferroelectric layer **104** (not shown) can be transmitted to these wave computing devices. Alternatively, sound wave signals from these wave computing devices can interfere with each other in the wave guide **106**, and can be transduced to an electrical wave signal by the ferroelectric layer **104** (not shown).

(36) In some of these embodiments, as shown in FIG. **2**B, the wave guide **106** has protruding portions **106**P**1**, **106**P**2** laterally protruding from opposite sides of the stacking structure including the electrode **102***a* and the underlying ferroelectric layer **104** (not shown), and may extend along, for example, the direction X. In these embodiments, the wave guide **106** is formed in a line shape.

(37) Referring to FIG. **2**C, in some embodiments, the wave guide **106** fans out from two sides of the stacking structure including the electrode **102***a* and the underlying ferroelectric layer **104** (not shown). One protruding portion of the wave guide **106** (referred to as a protruding portion **106**P**3**) extends along the direction X from a first side of the stacking structure, while the other protruding portion of the wave guide **106** (referred to as a protruding portion **106**P**4**) extends along a direction Y from a second side of the stacking structure. The first side is adjacent to the second side, and the direction X is intersected with the direction Y. In these embodiments, the wave guide **106** may be formed in an “L” shape.

(38) Referring to FIG. **2**D, in some embodiments, the wave guide **106** fans out from three sides of

the stacking structure including the electrode **102***a* and the underlying ferroelectric layer **104** (not shown). Protruding portions **106**P**5**, **106**P**6** extends away from opposite first and second sides of the stacking structure, and a protruding portion **106**P**7** extends from a third side of the stacking structure adjacent to the first and second sides. As an example, the protruding portions **106**P**5**, **106**P**6** may extend along the direction X, and the protruding portion **106**P**7** may extend along the direction Y. In these embodiments, the wave guide **106** may be formed in a "T" shape.
(39) In some embodiments, the wave guide **106** has multiple protruding portions, and at least one of these protruding portions can have one or more bend(s). As an example shown in FIG. **2**E, the wave guide **106** has two protruding portions **106**P**8**, **106**P**9** laterally extending from two adjacent sides of the stacking structure including the electrode **102***a* and the underlying ferroelectric layer **104** (not shown). The protruding portion **106**P**8** may extend along the direction X. On the other hand, the protruding portion **106**P**9** may extend along the direction Y from the stacking structure, and may turn to the direction X at a distance from the stacking structure. It should be understood that the wave guide **106** described with reference to FIG. **2**A through FIG. **2**D may be otherwise formed with one or more protruding portion(s) having at least one bend.
(40) According to the above embodiments, the ferroelectric device **100** is a transducer that can transduce an electrical wave signal to a sound wave signal, or vice versa. According to a pre-programmed polarization state, the ferroelectric device **100** can pass or invert the incoming wave signal. In other embodiments that will be described in greater details, a ferroelectric device similar to the ferroelectric device **100** can perform transduction between an electrical wave signal and a spin wave signal or a plasmon wave signal.
(41) FIG. **3** is a schematic cross-sectional view illustrating a ferroelectric device **300**, according to some embodiments of the present disclosure. The ferroelectric device **300** is similar to the ferroelectric device **100** described above, thus only differences between the ferroelectric devices **100**, **300** will be described. The same or the like parts in the ferroelectric devices **100**, **300** would not be repeated again.
(42) Referring the FIG. **3**, according to some embodiments, the ferroelectric device **300** further includes a magnetostrictive layer **302** lying between the ferroelectric layer **104** and a wave guide **304**. When the ferroelectric device **300** is functioned as an input actuator, a lattice constant of the ferroelectric layer **104** changes with varying electrical field resulted from an electrical wave signal EW provided to the ferroelectric layer **104** through the electrode **102***a*. The change in lattice constant of the ferroelectric layer **104** is transduced to a change in magnetic anisotropy in the magnetostrictive layer **302**, which produces a magnetic wave signal MW (or referred to as a spin wave signal) in the wave guide **304**. In other words, an electrical energy is transduced to a mechanical energy via the ferroelectric layer **104**, and the mechanical energy is further transduced to a magnetic energy via the magnetostrictive layer **302**. On the other hand, when the ferroelectric device **300** is functioned as an output sensor, a magnetic wave signal MW provided to the wave guide **304** is transduced to an electrical wave signal EW by the magnetostrictive layer **302** and the ferroelectric layer **104**. The magnetic wave MW may cause mechanical strain in the magnetostrictive layer **302**, and such mechanical strain may stress the ferroelectric layer **104**. As a result of piezoelectricity (i.e., direct piezoelectricity) of the ferroelectric layer **104**, the ferroelectric layer **104** can induce an electric field while being stressed by the magnetostrictive layer **302**, and a resulting electrical wave signal (e.g., the electrical wave signal EW) is output by the electrode **102***a*. In either way of transduction, the magnetic wave MW can be in phase or out of phase with the electrical wave signal EW, depending on the polarization state pre-programmed in the ferroelectric layer **104**.
(43) In these embodiments, the magnetostrictive layer **302** is formed of a magnetostrictive material. As examples, the magnetostrictive material may include CoFeB or FeGa. Further, in these embodiments, the wave guide **304** is formed of a magnetic material. For instance, the wave guide **304** may be formed of CoFeB, CoFe, NiFe, GaFe, CoFeO or the like. Further, as similar to the

wave guide **106** described with reference to FIG. **2**A through FIG. **2**E, the wave guide **304** may laterally protrude from one or more sides of a stacking structure including the electrode **102***a*, the ferroelectric layer **104** and the magnetostrictive layer **302**, and the protruding portion(s) of the wave guide **304** may or may not have bend(s).

(44) FIG. **4**A is a schematic cross-sectional view illustrating a ferroelectric device **400***a*, according to some embodiments of the present disclosure. The ferroelectric device **400***a* is similar to the ferroelectric device **100** described above, thus only differences between the ferroelectric devices **100**, **400***a* will be described. The same or the like parts in the ferroelectric devices **100**, **400***a* would not be repeated again.

(45) Referring to FIG. **4**A, according to some embodiments, the ferroelectric layer **104** is in contact with a wave guide **402** through a piezoelectric layer **404**. As a difference from the ferroelectric layer **104**, the piezoelectric layer **404** may only possess piezoelectricity (direct and inverse piezoelectricity), rather than possessing both of piezoelectricity and ferroelectricity. In other words, the piezoelectric layer **404** may not be able to be programmed with different polarization states (e.g., the polarization states P**1**, P**2** as described with reference to FIG. **1**A and FIG. **1**B). When the ferroelectric device **400***a* is functioned as an input actuator, an electrical wave signal EW provided to the electrode **102***a* may be transduced to varying mechanical stress by inverse piezoelectricity of the ferroelectric layer **104**, and the varying mechanical stress may be transduced back to an electrical wave signal (or referred to as a plasmon wave signal PW) by direct piezoelectricity of the piezoelectric layer **404**. The resulted plasmon wave signal PW can be further directed through the wave guide **402**. When the ferroelectric device **400***a* is functioned as an output sensor, a plasmon wave signal PW provided to the piezoelectric layer **404** through the wave guide **402** results in change of lattice constant of the piezoelectric layer **404**, thus the ferroelectric layer **104** is stressed by the piezoelectric layer **404**, so as to produce an electrical wave signal EW. In either case, an electrical energy is transduced to a mechanical energy, then transduced back to an electrical energy. Further, the plasmon wave signal PW can be in phase or out of phase with the electrical wave signal EW, depending on the polarization state pre-programmed in the ferroelectric layer **104**.

(46) In some embodiments, the piezoelectric layer **404** has low electrical conductivity, while the wave guide **402** has an electrical conductivity great enough for transmitting the plasmon wave PW. As an example, the piezoelectric layer **404** may be formed of an insulating lead zirconate titanate (PZT), while the wave guide **402** is formed of a conductive material, such as metal. Further, in these embodiments, the piezoelectric layer **404** and the ferroelectric layer **104** may be stacked between the electrodes **102***a*, **102***b*, and the wave guide **402** may be in lateral contact with the piezoelectric layer **404**. Although only a single wave guide **402** is depicted, more wave guides **402** may be in lateral contact with multiple sides of the piezoelectric layer **404**, and each of these wave guides **402** may or may not have bend(s), as similar to the wave guide **106** described with reference to FIG. **2**A through FIG. **2**E.

(47) FIG. **4**B is a schematic cross-sectional view illustrating a ferroelectric device **400***b*, according to some embodiments of the present disclosure. The ferroelectric device **400***b* is similar to the ferroelectric device **400***a* described with reference to FIG. **4**A, thus only differences between the ferroelectric devices **400***a*, **400***b* will be described. The same or the like parts in the ferroelectric devices **400***a*, **400***b* would not be repeated again.

(48) Referring to FIG. **4**B, in some embodiments, a wave guide **406** is in contact with the ferroelectric layer **104** without an additional piezoelectric layer in between. In these embodiments, the wave guide **406** is electrically conductive, and is formed of a piezoelectric material only possessing piezoelectricity, rather than both of ferroelectricity and piezoelectricity. In addition to transmitting a plasmon wave PW, the wave guide **406** is further configured to transduce a varying mechanical stress to the plasmon wave PW, or vice versa. When the ferroelectric device **400***b* is functioned as an input actuator, an electrical wave signal EW provided to the ferroelectric layer **104** is transduced to a varying mechanical stress by inverse piezoelectricity of the ferroelectric layer

104, and then the varying mechanical stress is transduced to a plasmon wave signal PW by direct piezoelectricity of the wave guide **406**. On the other hand, when the ferroelectric device **400***b* is functioned as an output sensor, a plasmon wave PW provided to the wave guide **406** is transduced to a varying mechanical stress by inverse piezoelectricity of the wave guide **406**, and the varying mechanical stress is then transduced to an electrical wave signal EW by direct piezoelectricity of the ferroelectric layer **104**. In either case, the plasmon wave signal PW can be in phase or out of phase with the electrical wave signal EW, depending on the polarization state pre-programmed in the ferroelectric layer **104**.

(49) In order to transmit the plasmon wave PW, a piezoelectric material for forming the wave guide **406** should be electrically conductive. As examples, the wave guide **406** may be formed of InN, GaN or AN. Further, as similar to the wave guide **106** described with reference to FIG. **2**A through FIG. **2**E, the wave guide **406** may laterally protrude from one or more sides of a stacking structure including the electrode **102***a* and the ferroelectric layer **104**, and the protruding portion(s) of the wave guide **304** may or may not have bend(s).

(50) FIG. **4**C is a schematic cross-sectional view illustrating a ferroelectric device **400***a′*, according to some embodiments of the present disclosure. The ferroelectric device **400***a′* is similar to the ferroelectric device **400***a* described with reference to FIG. **4**A, thus only differences between the ferroelectric devices **400***a*, **400***a′* will be described. The same or the like parts in the ferroelectric devices **400***a*, **400***a′* would not be repeated again.

(51) Referring to FIG. **4**C, in some embodiments, the ferroelectric layer **104** is sandwiched between the electrode **102***a* and an electrode **102***b′*, and the electrode **102***b′* lies in between the ferroelectric layer **104** and the piezoelectric layer **404**. The electrode **102***a* is configured to receive or output an electrical wave signal EW, whereas the electrode **102***b′* is coupled to a reference voltage, such as a ground voltage. In these embodiments, a varying mechanical stress produced by the ferroelectric layer **104** may stress the piezoelectric layer **404** through the electrode **102***b′* when the ferroelectric device **400***a′* is functioned as an input actuator, and a varying mechanical stress produced by the piezoelectric layer **404** may stress the ferroelectric layer **104** through the electrode **102***b′* when the ferroelectric device **400***a′* is functioned as an output sensor. Particularly, when the ferroelectric device **400***a′* is operated as an input actuator, the electrical wave signal EW may be directed to the reference voltage terminal along a path P**1**, without reaching the piezoelectric layer **404**. Accordingly, the transduction at the piezoelectric layer **404** may be solely affected by the varying mechanical stress produced by the ferroelectric layer **104**, rather than being undesirably affected by both of the varying mechanical stress produced by the ferroelectric layer **104** and the electrical wave signal EW. Similarly, when the ferroelectric device **400***a′* is functioned as an output sensor, the plasmon wave signal PW is directed to the reference voltage terminal along a path P**2**, without reaching the ferroelectric layer **104**. Therefore, the transduction at the ferroelectric layer **104** may be solely affected by the varying mechanical stress produced by the piezoelectric layer **404**.

(52) FIG. **4**D is a schematic cross-sectional view illustrating a ferroelectric device **400***b′*, according to some embodiments of the present disclosure. The ferroelectric device **400***b′* is similar to the ferroelectric device **400***b* described with reference to FIG. **4**B, thus only differences between the ferroelectric devices **400***b*, **400***b′* will be described. The same or the like parts in the ferroelectric devices **400***b*, **400***b′* would not be repeated again.

(53) Referring to FIG. **4**D, in some embodiments, the ferroelectric layer **104** is sandwiched between the electrode **102***a* and an electrode **102***b′*, and the electrode **102***b′* lies between the ferroelectric layer **104** and the wave guide **406**. The electrode **102***a* is configured to receive or output an electrical wave signal EW, whereas the electrode **102***b′* is coupled to a reference voltage, such as a ground voltage. In these embodiments, a varying mechanical stress produced by the ferroelectric layer **104** may stress the wave guide **406** through the electrode **102***b′* when the ferroelectric device **400***b′* is functioned as an input actuator, and a varying mechanical stress produced by the wave guide **406** may stress the ferroelectric layer **104** through the electrode **102***b′*

when the ferroelectric device **400***b′* is functioned as an output sensor. Particularly, when the ferroelectric device **400***b′* is operated as an input actuator, the electrical wave signal EW may be directed to the reference voltage terminal along a path P**1**, without reaching the wave guide **406**. Accordingly, the transduction at the wave guide **406** may be solely affected by the varying mechanical stress produced by the ferroelectric layer **104**, rather than being undesirably affected by both of the varying mechanical stress produced by the ferroelectric layer **104** and the electrical wave signal EW. Similarly, when the ferroelectric device **400***b′* is functioned as an output sensor, the plasmon wave signal PW is directed to the reference voltage terminal along a path P**2**, without reaching the ferroelectric layer **104**. Therefore, the transduction at the ferroelectric layer **104** may be solely affected by the varying mechanical stress produced by the wave guide **406**.

(54) According to various embodiments described above, the ferroelectric device can be functioned a transducer (input actuator or output sensor), and is able to pass or invert an incoming signal, based on the polarization state pre-programmed into the ferroelectric device. In addition, a plurality of the ferroelectric devices as input actuators can be connected to other wave computing logic devices as well as a plurality of the ferroelectric device as output sensors, to form a wave computing circuit, and the wave computing circuit may be further connected to a CMOS logic circuit or an electric memory. In some embodiments that will be described in greater details, a wave computing logic device includes one or more of the ferroelectric devices as input actuators or an output sensor.

(55) FIG. **5**A is a schematic cross-sectional view illustrating a wave computing logic device **500**, according to some embodiments of the present disclosure.

(56) Referring to FIG. **3** and FIG. **5**A, a plurality of the ferroelectric devices **300** as shown in FIG. **3** (e.g., 3 ferroelectric devices **300**) and an output sensor **502** are interconnected, to form the wave computing logic device **500** as shown in FIG. **5**A. The ferroelectric devices **300** may be connected via a shared wave guide **304**. In addition, the wave guide **304** may further connect the ferroelectric devices **300** to the output sensor **502**. As similar to each of the ferroelectric devices **300**, the output sensor **502** includes a pair of electrodes **102***a*, **102***b* at opposite sides of the wave guide **304**, and includes a magnetostrictive layer **302** lying between the wave guide **304** and the electrode **102***a*. As a difference from the ferroelectric devices **300**, the magnetostrictive layer **302** and the electrode **102***a* are in contact with each other through a piezoelectric layer **504**. The piezoelectric layer **504** may only possess piezoelectricity, rather than possessing both of ferroelectricity and piezoelectricity. In other words, the piezoelectric layer **504** may not be able to invert a wave signal, and may not be pre-programmed with a polarization state. In some embodiments, the piezoelectric layer **504** is formed of insulating lead zirconate titanate (PZT), InN, GaN or AN.

(57) The wave computing logic device **500** can be operated as a majority gate that produces a standing wave having the phase of the majority of the input electrical wave signals. As an example, inputs terminals IN**1**, IN**2**, IN**3** of the wave computing logic device **500** are respectively connected to the electrode **102***a* of one of the ferroelectric devices **300**, and are configured to receive the input electrical wave signals. These input electrical wave signals may be in phase with one another, or at least one of these input electrical wave signals may be out of phase with other input electrical wave signals. For instance, input electrical wave signals EW**1** in phase with each other are provided to the input terminals IN**1**, IN**2**, whereas an input electrical wave signal EW**2** out of phase with respect to the input electrical wave signals EW**1** is provided to the input terminal IN**3**. Further, the ferroelectric layers **104** of the ferroelectric devices **300** are each programmed with the same polarization state P**1**. Accordingly, the input electrical wave signals EW**1**, EW**2** are transduced to magnetic waves through the ferroelectric layers **104** having the polarization state P**1** and the magnetostrictive layers **302**, and the magnetic waves are in phase with the input electrical wave signals EW**1**, EW**2**, respectively. Further, these magnetic waves interfere with one another in the wave guide **304**, to form a standing wave signal SMW traveling along the wave guide **304**. The magnetic waves transduced from the input electrical wave signal EW**2** and one of the input

electrical wave signal EW**1** may cancel each other, such that the standing wave signal SMW is in phase with the magnetic wave transduced from the other input electrical wave signal EW**1**. As a result of the destructive interference, the standing wave signal SMW has the phase of the majority of the input electrical wave signals (e.g., the input electrical wave signals EW**1**). In some embodiments, a spacing between adjacent ferroelectric devices **300** is an integer number of the wavelength of the waves produced by the ferroelectric devices **300**, in order to maximize the interference between these waves.

(58) The standing wave signal SMW may be transduced back to an electrical wave signal EW**3** through the magnetostrictive layer **302** and the piezoelectric layer **504** in the output sensor **502**, and the electrical wave signal EW**3** may be output from an output terminal OUT connected to the electrode **102***a* of the output sensor **502**. Since each of the magnetostrictive layer **302** and the piezoelectric layer **504** can only perform transduction (rather than doing both of transduction and inversion), the output electrical wave signal EW**3** is in phase with the standing wave signal SMW and the phase of the majority of the input electrical wave signals (e.g., the input electrical wave signals EW**1**).

(59) Although the wave computing logic device **500** is depicted as having three inputs, the wave computing logic device **500** may alternatively have more than three inputs. The present disclosure is not limited to an amount of the inputs of a wave computing logic device **500**. Moreover, the ferroelectric layers **104** at the inputs of the wave computing logic device **500** may be alternatively pre-programmed with different polarization states, as will be further described in details.

(60) FIG. **5**B is a schematic cross-sectional view illustrating the wave computing logic device **500** having the ferroelectric devices **300** pre-programmed with different polarization states P**1**, P**2**, according to some embodiments of the present disclosure.

(61) Referring to FIG. **5**B, as another example, the ferroelectric layers **104** at the inputs of the wave computing logic device **500** are programmed with different polarization states P**1**, P**2**. Specifically, the ferroelectric layers **104** of the ferroelectric devices **300** connected to the input terminals IN**2**, IN**3** are pre-programmed with the polarization state P**1**, whereas the ferroelectric layer **104** of the ferroelectric device **300** connected to the input terminal IN**1** is pre-programmed with the polarization state P**2**. In this way, the input electrical wave signal EW**1** provided to the ferroelectric device **300** connected to the input terminal IN**1** is transduced and inverted, whereas other input electrical wave signals EW**1**, EW**2** provided to the ferroelectric devices **300** connected to the input terminals IN**2**, IN**3** are merely transduced. Due to destructive interference at the wave guide **304**, the resulting standing wave signal SMW is in phase with the input electrical wave signal EW**2**. Therefore, the output electrical wave signal EW**3** transduced from the standing wave signal SMW is in phase with the input electrical wave signal EW**2** as well.

(62) The wave computing logic device **500** as shown in FIG. **5**B can be regarded as a combination of a majority gate and an inverter (at one of its inputs). By changing polarization states of the ferroelectric layers **104** at inputs of the wave computing logic device **500**, more combinations of input electrical wave signals and output electrical wave signal can be obtained. If the number of inputs is n, then the total number of reconfigurable gates is 2 to the power n. In further embodiments, as will be described, the ferroelectric device **300** can be used at an output of a wave computing logic device.

(63) FIG. **6**A is a schematic cross-sectional view illustrating a wave computing logic device **600**, according to some embodiments of the present disclosure.

(64) The wave computing logic device **600** can be functioned as a majority gate, as similar to the wave computing logic device **500** as shown in FIG. **5**A. As a difference from the wave computing logic device **500**, the wave computing logic device **600** includes the ferroelectric device **300** as an output sensor, and includes a plurality of input actuators **602** connected to the ferroelectric device **300** via a shared wave guide **304**. As similar to the ferroelectric device **300**, the input actuators **602** respectively include a pair of electrodes **102***a*, **102***b* at opposite sides of the wave guide **304**, and

include a magnetostrictive layer **302** lying between the wave guide **304** and the electrode **102***a*. In addition, the input actuator **602** further includes a piezoelectric layer **604** sandwiched between the electrode **102***a* and the piezoelectric layer **302**. The piezoelectric layer **604** may only possess piezoelectricity, rather than possessing both of ferroelectricity and piezoelectricity. In other words, as different from the ferroelectric layer **104** in the ferroelectric device **300**, the piezoelectric layer **604** in the input actuator **602** may not be able to invert a wave signal, and may not be pre-programmed with a polarization state. In some embodiments, the piezoelectric layer **604** is formed of insulating lead zirconate titanate (PZT), InN, GaN or AN.
(65) Input electrical wave signals provided to the input actuators **602** may be transduced by the input actuators **602** and interfered with one another to form a standing wave signal in the wave guide **304**, then the standing wave signal is transduced to an output electrical wave signal by the ferroelectric device **300**. Further, the output electrical wave signal can be in phase or out of phase with the standing wave signal, depending on the polarization state of the ferroelectric layer **104** in the ferroelectric device **300**. As an example, the wave computing logic device **600** includes 3 input actuators **602**, and input terminals IN**1**, IN**2**, IN**3** of the wave computing logic device **600** are respectively connected to the electrode **102***a* of one of the input actuators **602** and configured to receive input electrical wave signals. These input electrical wave signals may be in phase with one another, or at least one of these input electrical wave signal may be out of phase with other input electrical wave signals. For instance, input electrical wave signals EW**1** in phase with each other are provided to the input terminals IN**1**, IN**2**, whereas an input electrical wave signal EW**2** out of phase with respect to the input electrical wave signals EW**1** is provided to the input terminal IN**3**. The input electrical wave signals EW**1**, EW**2** are transduced to magnetic wave signals by the piezoelectric layers **604** and the magnetostrictive layers **302** in the input actuators **602**, without being inversed. Further, the magnetic wave signals interfere with each other in the wave guide **304**, and the resulting standing wave signal SMW having the phase of the majority of the input electrical wave signals EW**1**, EW**2** travels along the wave guide **304**. The magnetostrictive layer **302** and the ferroelectric layer **104** in the ferroelectric device **300** transduce the standing wave signal SMW to an output electrical wave signal EW**3**. As an example illustrated in FIG. **6**A, the ferroelectric layer **104** is pre-programmed with the polarization state P**1**, and the output electrical wave signal EW**3** is in phase with the standing wave signal SMW, which should be the phase of the majority of the input electrical wave signals EW**1**, EW**2**. In some embodiments, a spacing between adjacent input actuators **602** is an integer number of the wavelength of the waves produced by the input actuators **602**, in order to maximize the interference between these waves.
(66) FIG. **6**B is a schematic cross-sectional view illustrating the wave computing logic device **600** having the ferroelectric device **300** pre-programmed with the polarization state P**2**, according to some embodiments of the present disclosure.
(67) Referring to FIG. **6**B, as another example, the ferroelectric layer **104** in the ferroelectric device **300** of the wave computing logic device **600** is programmed with the polarization state P**2**. Accordingly, the standing wave signal SMW is transduced and inverted by the ferroelectric device **300**, thus the resulting output electrical wave signal EW**3** is out of phase with the majority of the input electrical wave signals EW**1**, EW**2**. Therefore, the wave computing logic device **600** as shown in FIG. **6**B can be regarded as a combination of a majority gate and an inverter (at its output).
(68) It should be understood that, the magnetostrictive layers **302** shown in FIG. **5**A, FIG. **5**B, FIG. **6**A and FIG. **6**B may be omitted, such that the standing wave signal resulted in the wave guide **304** should be a sound wave signal. Alternatively, the magnetostrictive layers **302** in the ferroelectric devices **300** shown in FIG. **5**A, FIG. **5**B may be replaced by piezoelectric layers, and the magnetostrictive layer **302** in the output sensor **502** shown in FIG. **5**A and FIG. **5**B may be omitted. Similarly, the magnetostrictive layers **302** in the input actuators **602** shown in FIG. **6**A and FIG. **6**B may be replaced by piezoelectric layers, and the magnetostrictive layer **302** in the ferroelectric

device **300** shown in FIG. **6**A and FIG. **6**B may be omitted. In these alternative embodiments, the standing wave signal resulted in the wave guide **304** should be a plasmon wave.
(69) In addition to be used in a wave computing logic device, the ferroelectric device according to various embodiments may be used to generate coherent differential wave signals.
(70) FIG. **7**A is a schematic plan view illustrating a differential wave signal generator **700**, according to some embodiments of the present disclosure. FIG. **7**B is a schematic cross-sectional view along X-X′ line shown in FIG. **7**A.
(71) Referring to FIG. **7**A and FIG. **7**B, the differential wave signal generator **700** includes two of the ferroelectric devices **100** as shown in FIG. **1**A and FIG. **1**B. The ferroelectric devices **100** may be connected by a shared electrode **102***b*, while the electrode **102***a*, the ferroelectric layer **104** and the wave guide **106** in one of the ferroelectric device **100** are laterally spaced apart from the electrode **102***a*, the ferroelectric layer **104** and the wave guide **106** in the other ferroelectric device **100**. In some embodiments, the wave guides **106** may be substantially parallel with each other. Further, the ferroelectric layer **104** in one of the ferroelectric devices **100** is pre-programmed with the polarization state P**1**, while the ferroelectric layer **104** in the other ferroelectric device **100** is pre-programmed with the polarization state P**2**.
(72) The electrodes **102***a* of the ferroelectric devices **100** may be configured to receive identical electrical wave signals. One of the electrical wave signals is transduced to a sound wave signal in phase with the electrical wave signals through the ferroelectric layer **104** pre-programmed with the polarization state P**1**, whereas the other electrical wave signal is transduced to a sound wave signal out of phase with the electrical wave signals through the ferroelectric layer **104** pre-programmed with the polarization state P**2**. In other words, two sound wave signals are generated which are out of phase. As a result, a pair of differential wave signals can be output via the wave guides **106**.
(73) It should be understood that, when both ferroelectric layers **104** are pre-programmed with the same polarization state, sound wave signals can be generated along the wave guides **106** which are in phase. Further, in other embodiments, a magnetostrictive layer or a piezoelectric layer may be further inserted in between each ferroelectric layer **104** and the contacted wave guide **106**, such that the resulting wave signals can be magnetic wave signals or plasmon wave signals.
(74) FIG. **8** is a schematic diagram illustrating that the differential wave signal generator **700** as shown in FIG. **7** is further connected to a wave computing logic device **800**, according to some embodiments of the present disclosure.
(75) Referring to FIG. **8**, a pair of differential wave signals WS**1**, WS**2** are generated along the wave guides **106** of the differential wave signal generator **700** as shown in FIG. **7**. One of the wave guides **106** is further connected to a wave computing logic device **800**, and the wave signal traveled along this wave guide **106** (e.g., the wave signal WS**1**) is provided as an input of a logic operation performed by the wave computing logic device **800**. Correspondingly, the wave computing logic device **800** may output a wave signal WS**3** along a wave guide **802**. By comparing the wave signal WS**3** with the wave signal WS**2**, the phase of the wave signal WS**3** can be identified. The wave signal WS**3** may be either in phase with the wave signal WS**2**, or out of phase with the wave signal WS**2**. As an example, the wave computing logic device **800** may be an inverter, such that the wave signal WS**3** may be out of phase with respect to the wave signal WS**1**, and may be in phase with the wave signal WS**2**. As another example, the wave computing logic device **800** is a majority gate, or a combination of majority gate and an inverter (such as the wave computing logic devices **500**, **600** as described with reference to FIG. **5**A, FIG. **5**B, FIG. **6**A and FIG. **6**B), and more input wave signals may be provided to the wave computing logic device **800**.
(76) FIG. **9** is a schematic diagram illustrating an error correction device **900** used in a wave computing circuit, according to some embodiments of the present disclosure.
(77) Referring to FIG. **9**, the error correction device **900** may include three of the ferroelectric devices **100** as shown in FIG. **1**A and FIG. **1**B. The ferroelectric devices **100** may be connected by a shared electrode **102***b*, while the electrode **102***a*, the ferroelectric layer **104** and the wave guide

106 in each of the ferroelectric devices **100** are laterally spaced apart from the electrode **102***a*, the ferroelectric layer **104** and the wave guide **106** in another one of the ferroelectric device **100**. In some embodiments, the wave guides **106** may be substantially parallel with one another. Further, the ferroelectric layers **104** in two of the ferroelectric devices **100** are pre-programmed with the polarization state P**1**, while the ferroelectric layer **104** in the other ferroelectric device **100** is pre-programmed with the polarization state P**2**.
(78) The electrodes **102***a* of the ferroelectric devices **100** may be configured to receive identical electrical wave signals. Two of the electrical wave signals are transduced to wave signals WS**1**, which are in phase with the input electrical wave signals, by the ferroelectric layers **104** pre-programmed with the polarization state P**1**. The other electric wave signal is transduced to a wave signal WS**2** which is out of phase with the input electrical wave signals, by the ferroelectric layer **104** pre-programmed with the polarization state P**2**.
(79) A wave computing logic device **902** may be connected to one of the wave guides **105** along which one of the wave signals WS**1** travels, such that this wave signal WS**1** is provided as an input of a logic operation performed by the wave computing logic device **902**. Correspondingly, the wave computing logic device **902** outputs a wave signal WS**3** along a wave guide **904**. Ideally, the wave signal WS**3** is either in phase with the wave signals WS**1**, or with the wave signal WS**2**. However, in certain cases, the wave signal WS**3** may lose coherence with the wave signals WS**1**, WS**2**, such that the wave signal WS**3** may have a phase shift greater than 0° and less than 180°, with respect to each of the wave signals WS**1**, WS**2**. In order to correct this error, the wave signal WS**3** is compared with each of the wave signals WS**1**, WS**2**, to determine the phase of the wave signal WS**3** is closer to which of the phase of the wave signal WS**1** and the phase of the wave signal WS**2**. According to the comparison result, a corrected wave signal (in phase with one of the wave signals WS**1**, WS**2**) can be output. According to some embodiments, a majority gate (e.g., the majority gate as described with reference to FIG. **5**A) can be used for comparing the wave signals WS**1**, WS**2**, WS**3**.
(80) It should be understood that, each of the ferroelectric devices **100** in the error correction device **900** can be replaced by the ferroelectric device **300** as shown in FIG. **3**, the ferroelectric device **400***a* shown in FIG. **4**A, the ferroelectric device **400***b* shown in FIG. **4**B, the ferroelectric device **400***a*′ shown in FIG. **4**C or the ferroelectric device **400***b*′ shown in FIG. **4**D.
(81) The ferroelectric devices used in various wave computing devices as described above can be formed in a semiconductor die. In some embodiments, the wave computing components are integrated with CMOS circuits in the same semiconductor die.
(82) FIG. **10** is a schematic cross-sectional view illustrating a semiconductor die **1000**, according to some embodiments of the present disclosure.
(83) Referring to FIG. **10**, the semiconductor die **1000** is built on a semiconductor substrate **1002**. A front-end-of-line (FEOL) structure FL in the semiconductor die **1000** may include a plurality of CMOS transistors formed on the semiconductor substrate **1002**. According to some embodiments, these CMOS transistors have channel structures **1004** standing on the semiconductor substrate **1002**, and include gate structures **1006** covering and intersected with the channel structures **1004**. In addition to the FEOL structure FL, the semiconductor die **1000** also includes a back-end-of-line (BEOL) structure BL formed on the FEOL structure FL. The BEOL structure BL may include a stack of dielectric layers **1008** formed on the semiconductor substrate **1002** and covering the FEOL structure FL, and include multiple metallization tiers Mn embedded in the stack of dielectric layers **1008**. Bottommost metallization layer(s) Mn may provide local interconnections for the CMOS transistors, while upper metallization layers Mn may provide global interconnections.
(84) Moreover, a wave computing circuit **1010** is also embedded in the BEOL structure BL. The wave computing circuit **1010** is formed in the stack of dielectric layers **1008**. In some embodiments, the wave computing circuit **1010** is formed over several metallization tiers Mn. Although not shown, more metallization tiers may be further disposed on the wave computing

circuit **1010**. The wave computing circuit **1010** may include input actuators, output sensors and various wave computing logic devices, and may also include differential wave signal generators and/or error correction devices. As described above, the ferroelectric device according to various embodiments (e.g., the ferroelectric device **100**) can be used in each of the input actuators, output sensors, wave computing logic devices, the differential signal generators and error correction devices. Further, the wave computing circuit **1010** may be connected to the CMOS transistors in the FEOL structure FL through the metallization tiers Mn lying in between. Therefore, computing in the semiconductor die **1000** may be executed by a combination of a CMOS circuit including the CMOS transistors and the wave computing circuit **1010**.

(85) As above, a ferroelectric device used in a wave computing circuit is provided. The ferroelectric device includes a ferroelectric layer configured to transduce an electrical wave signal to a varying mechanical stress by inverse piezoelectricity, and the varying mechanical stress may result in a sound wave, a plasmon wave or a magnetic wave. Alternatively, the ferroelectric layer can transduce a varying mechanical stress resulted from a sound wave, a plasmon wave or a magnetic wave, to an electrical wave signal by direct piezoelectricity. Further, the output wave signal can be in phase, or 180° out of phase with respect to the input wave signal, depending on a polarization state pre-programmed in the ferroelectric layer. In other words, the ferroelectric device can transduce a wave signal without inverting the wave signal, or can transduce and invert a wave signal. Further, the inversion of wave signal can be performed right at the ferroelectric device, without traveling along a delay line or interfering with a reference wave signal. Therefore, the ferroelectric device is able to invert a wave signal on the spot. According to some embodiments, the ferroelectric device is functioned as a wave transducer, which could be further capable of inverting a wave signal. In other embodiments, the ferroelectric device can be used in a majority gate, a differential wave signal generator, an error correction device or another wave computing device.

(86) In an aspect of the present disclosure, a ferroelectric device is provided. The ferroelectric device comprises: first and second electrodes; a ferroelectric layer, disposed between the first and second electrodes, and configured to transduce an electrical wave signal to a varying mechanical stress by piezoelectricity, or vice versa, wherein a first polarization state or a second polarization state opposite to the first polarization state is programmed in the ferroelectric layer; and a wave guide, in contact with the ferroelectric layer, and configured to transmit a wave signal resulted from or resulting the varying mechanical stress, wherein the wave signal is in phase with the electrical wave signal when the ferroelectric layer is programmed with the first polarization state, and the wave signal is coherent and out of phase with the electrical wave signal when the ferroelectric layer is programmed with the second polarization state.

(87) In another aspect of the present disclosure, a wave computing device is provided. The wave computing device comprises: a ferroelectric device and a wave transducer connected to each other through a wave guide, wherein the ferroelectric device comprises: first and second electrodes; and a ferroelectric layer, in contact with the wave guide and located between the first and second electrodes, wherein the ferroelectric layer is configured to transduce an electrical wave signal to a varying mechanical stress by piezoelectricity, or vice versa, and wherein the ferroelectric layer is programmed with a first polarization state or a second polarization state opposite to the first polarization state.

(88) In yet another aspect of the present disclosure, a wave computing device is provided. The wave computing device comprises: a first ferroelectric layer and a second ferroelectric layer, laterally separated from each other wherein the first ferroelectric layer is programmed with a first polarization state, and the second ferroelectric layer is programmed with a second polarization state opposite to the first polarization state; a first wave guide and a second wave guide, wherein the first wave guide is in contact with the first ferroelectric layer, and the second wave guide is in contact with the second ferroelectric layer; a first electrode and a second electrode, configured to receive identical electrical wave signals, wherein the first ferroelectric layer lies between the first electrode

and the first wave guide, and the second ferroelectric layer lies between the second electrode and the second wave guide; and a common electrode, disposed at a side of the first and second wave guides facing away from the first and second ferroelectric layers.
(89) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A ferroelectric device, comprising: first and second electrodes; a ferroelectric layer, disposed between the first and second electrodes, and configured to transduce an electrical wave signal to a varying mechanical stress by piezoelectricity, or vice versa, wherein a first polarization state or a second polarization state opposite to the first polarization state is programmed in the ferroelectric layer; and a wave guide, configured to transmit a wave signal resulted from or resulting the varying mechanical stress, wherein the wave signal is in phase with the electrical wave signal when the ferroelectric layer is programmed with the first polarization state, and the wave signal is out of phase with the electrical wave signal when the ferroelectric layer is programmed with the second polarization state.
2. The ferroelectric device according to claim 1, wherein the ferroelectric layer is in contact with the wave guide through a magnetostrictive layer.
3. The ferroelectric device according to claim 2, wherein the wave signal is a magnetic wave signal, and the wave guide is formed of a magnetic material.
4. The ferroelectric device according to claim 1, wherein the ferroelectric layer is in contact with the wave guide through a piezoelectric layer possessing piezoelectricity, rather than both of piezoelectricity and ferroelectricity.
5. The ferroelectric device according to claim 4, wherein the wave signal is a plasmon wave signal, and the wave guide is electrically conductive.
6. The ferroelectric device according to claim 1, wherein the wave guide is formed of a piezoelectric material, and is electrically conductive.
7. The ferroelectric device according to claim 1, wherein the ferroelectric layer and the wave guide extend between the first and second electrodes.
8. The ferroelectric device according to claim 1, wherein one of the first and second electrodes extends between the ferroelectric layer and the wave guide.
9. The ferroelectric device according to claim 1, wherein the wave guide laterally protrudes with respect to one or more sides of the ferroelectric layer.
10. The ferroelectric device according to claim 9, wherein a laterally protruding portion of the wave guide has at least one bend along its extending direction.
11. A wave computing device, comprising: a ferroelectric device and a wave transducer connected to each other through a wave guide, wherein the ferroelectric device comprises: first and second electrodes; and a ferroelectric layer, located between the first and second electrodes, wherein the ferroelectric layer is configured to transduce an electrical wave signal to a varying mechanical stress by piezoelectricity, or vice versa, and wherein the ferroelectric layer is programmed with a first polarization state or a second polarization state opposite to the first polarization state.
12. The wave computing device according to claim 11, wherein the ferroelectric device is an input actuator and the wave transducer is an output sensor, wherein the electrical wave signal is an input

electrical wave signal, and wherein the ferroelectric device transduces the input electric wave signal to a wave signal resulted from the varying mechanical stress.

13. The wave computing device according to claim 12, further comprising more of the ferroelectric devices connected to the wave transducer through the wave guide and producing additional wave signals based on additional input electrical wave signals.

14. The wave computing device according to claim 13, wherein the wave signal and the additional wave signals interfere with one another in the wave guide, to produce a standing wave signal traveling along the wave guide, and wherein the wave transducer translates the standing wave signal to an output electrical signal.

15. The wave computing device according to claim 14, wherein each of the standing wave signal and the output electrical wave signal has a phase of a majority of the input electrical wave signal and the additional input electrical wave signals.

16. The wave computing device according to claim 11, wherein the ferroelectric device is an output sensor and the wave transducer is an input actuator, wherein the electrical wave signal is an output electrical wave signal, and wherein the ferroelectric device transduces a standing wave signal traveling along the wave guide and resulting the varying mechanical stress, to the output electrical wave signal.

17. The wave computing device according to claim 16, further comprising more of the input actuators, wherein the input actuators are connected to the ferroelectric device through the wave guide, and produce wave signals based on input electrical wave signals, and wherein the wave signals interfere with one another to form the standing wave signal.

18. A wave computing device, comprising: a wave guide; a ferroelectric device, wherein the ferroelectric device comprises: a first electrode; a second electrodes; and a ferroelectric layer, wherein the first electrode is disposed at a first side of the wave guide, the second electrode and the ferroelectric layer are disposed at a second side of the wave guide, the second side is opposite to the first side, the ferroelectric layer is located between the wave guide and the second electrode, the ferroelectric layer is configured to transduce an electrical wave signal to a varying mechanical stress by piezoelectricity, or vice versa, and wherein the ferroelectric layer is programmed with a first polarization state or a second polarization state opposite to the first polarization state; and an output sensor connected to the ferroelectric device through the wave guide.

19. The wave computing device according to claim 18, wherein the ferroelectric device further comprises a magnetostrictive layer between the wave guide **304** and the second electrode.

20. The wave computing device according to claim 18, wherein the output sensor comprises a piezoelectric layer sandwiched by a pair of electrodes.